

INTERNATIONAL CONFERENCE ON RECENT TRENDS IN ADVANCED COMPUTING
2019, ICRTAC 2019**MLPPT-MHS: Multi-Layered Privacy Preserving and Traceable
Mobile Health System**A.Divya Preetha^a, T.S.Pradeep Kumar^{a*}^aVIT University, Chennai 600127, India

Abstract

Personal Health Record (PHR) is a transpiring patient-centric embodiment of patient information trade that is customarily contracted out to third parties, for illustration, cloud service providers. There were far-reaching privacy apprehensions as individual medical records may possibly be unveiled to some third-party servers in addition to unconstitutional parties. On the road to warrant the patients' charge to access their medical records, it is a hopeful procedure to encrypt the medical records prior to outsourcing. Hitherto, questions like exposure to privacy, scalability during key management, in addition to proficient user revocation, have lingered the most imperative tests in the direction of accomplishing fine-grained access control. This manuscript puts forward a fresh multi-layer patient-centric scaffold in addition to an anthology of techniques in support of data access control to medical records amassed in honest-but-curious servers. In the direction of accomplishing fine-grained as well as scalable access control, a new-fangled modification of Cipher-text Policy Attribute-Based Encryption (CP-ABE) technique is persuaded with random key length in the offline encryption phase to encrypt each patient's PHR file. In the online phase, the CP-ABE encrypted data is transmitted to the cloud through a secure tunnel and then the data in rest is encrypted again by the 256-bit Advanced Encryption Standard (AES-256). In order to eliminate the problem aroused from leaked keys, both attribute and user revocation is done. The proposed system achieves data confidentiality, access control, user revocation, random key length, resists collusion attack, tracks the user location periodically, and preserves forward and backward secrecy. Within the MLPPT-MHS scheme, the computation time to perform Key Generation, Encryption, and Decryption would be 823.75ms, 152.43ms, 82.54ms and the storage overhead incurred to record the Public Parameters, Secret Key and Cipher Text would be 6098 bits, 59,653 bits, and 26,712 bits. These investigational outcomes demonstrate the security, scalability, as well as efficiency of this MLPPT-MHS scheme.

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* Corresponding author. Tel.: +91-999-462-6196.E-mail address: tspradeepkumar@vit.ac.in

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1. Introduction

In the current days, Personal Health Record (PHR) is materialized the same as a patient-centric representation of health data trade. A PHR amenity consents a patient on the way to generate, administer, over and above organize his/her individual health data at a single stop all the way in the course of the web that has prepared the storage, recuperation, in addition to medical data sharing as more proficient. Of late, several styles for amassing PHRs in the cloud were proposed in [1]-[3]. Despite the fact that it is invigorating to have expedient PHR amenities for all and sundry, there is numerous security as well as privacy menace that might hamper its ample espousal. Even though there is an assortment of healthcare directives like HIPAA which is lately adjusted to integrate commerce acquaintances [4]-[9]. Within this manuscript, a venture is made in the direction of gaining knowledge of the patient-centric, safe and sound PHR sharing, and contemplated on dealing with the intricate and tricky key management concerns. To amalgamate CP-ABE into a colossal PHR system, noteworthy apprehensions like scalability in key management, vibrant policy renewal, along with proficient on-demand revocation would be substantial to work out as well as stay behind principally unbolt till date. Till now, the authors compose the subsequent major benefactions:

1. The MLPPT-MHS scheme recommends a fresh CP-ABE-based skeleton for patient-centric safe and sound health data sharing in cloud atmosphere. To trim down the concerns acquired from lightweight encryption technique, the encryption process is conceptually divided as offline and online. Specifically, the patient data are encrypted using a fresh alternative of Cipher-text Policy Attribute-Based Encryption with random key length in the former, while the cipher-text obtained from the offline phase is transmitted through a secure tunnel to the cloud and then the encrypted data at rest is again encrypted using 256-bit Advanced Encryption Standard. In this fashion, the scaffold can concurrently grip dissimilar categories of PHR sharing applications' obligations, at the same time as acquiring negligible key administration overhead for the proprietors along with consumers in the scheme. Into the bargain, the skeleton makes obligatory write access control, leverages vibrant strategy modernization, and offers concise access to PHRs under crisis.

2. The activities of every user are monitored periodically using the session tracking technique. Key revocation is done along with user revocation in order to avoid conspirators from mishandling the leaked keys. Likewise forward and backward secrecy of patient data is achieved.

3. The MLPPT-MHT scheme endow with a meticulous investigation of the intricacy and efficiency of the anticipated safe and sound PHR sharing elucidation, with orientation to manifold metrics in computation, announcement, storage, as well as key management. In addition, the authors weigh their proposal against more than a few preceding ones in complexity, efficiency, as well as security. Moreover, the MLPPT-MHS scheme reveals its effectiveness by executing it on a contemporary workstation furthermore carrying out experimentations/replications.

2. Related Work

This manuscript is typically unified to efforts in cryptographically inflicted access control in favor of subcontracted medical records in addition to attribute-based encryption. On the road to comprehending access control, public key encryption methods [8]-[12] acquire elevated key management operating cost.

2.1 Facilitating Fine-Grained Access Control

ABE appears in two zests entitled key-policy ABE along with cipher-text-policy ABE. In the former technique, the attributes are exploited to portray the encrypted data as well as policies are put up into consumer's keys; at the same time as in the latter, attributes are exploited to portray a user's record, furthermore the party that encrypts, resolves a strategy on top of who may possibly decrypt the records. The latter is further pertinent to data outsourcing construction than the former for the reason that it facilitates data possessors to opt an access structure on, as well as to encrypt medical records that are ought to be outsourced by means of encrypting using the analogous communal attributes [13], [14]. Numerous works exploited ABE to apprehend fine-grained access control [15], [16], [9], [17]. More than ever, there is an escalating curiosity in pertaining ABE to safeguard electronic healthcare records (EHRs). In recent times, Narayan et al. propounded an attribute-based model in which each and every patient's details are encrypted via a new-fangled modification of CP-ABE [18] in addition to straight revocation. Conversely, the span of cipher-text nurtures linearly by means of the amount of unrevoked consumers. In [19], an alternative of Attribute Based Encryption that permits entrustment of access rights is anticipated. In [20], cipher-text policy ABE is pertained [21] to administer the distribution of medical records; moreover pioneered the perception of social or professional provinces. In [22], Akinyele et al. scrutinized by means of ABE to spawn self-protecting EMRs that know how to either be hoarded on cloud servers otherwise mobile phones with the intention that EMR may well be viewed once the provider is offline. Conversely, there are numerous widespread negative aspects of the previous exertions. Foremost, they habitually presume the exploitation of a trusted authority in the organization. This might produce a consignment bottleneck in addition to being ill with the key escrow crisis from the time when the Trust Authority would access every encrypted file, making way for probable privacy exposure. Into the bargain, it is not a matter-of-fact to entrust the entire attribute administration errands to a single TA, as well as officially stating all consumers' attributes or roles and engendering secret keys.

2.2 Revocable Attribute Based Encryption

This would be a renowned exigent crisis to rescind consumers/attributes proficiently as well as on-demand in ABE. Customarily, this is over and over again prepared via the authority broadcasting sporadic key updates headed for unrevoked consumers recurrently [15], [24], which does not pull off absolute backward/forward security; furthermore is not as much of proficient. Of late, [25] along with [26] planned two CP-ABE schemes in the company of instantaneous attribute revocation competence, in place of episodic revocation. To boot, Ruj et al. [27] proffered an unusual explanation to the equivalent problem using a dispersed ABE proposal [28]. The foremost enhancement of their explanation is, each and every user would be capable of getting hold of secret keys from whichever split of the trust authorities in the classification, in contradiction of the CC MAABE. LW ABE proposal achieves superior strategy eloquence; moreover it is expanded by [27] to prop up user revocation. The stumbling block is that the communication overhead caused by key revocation remains elevated, as it necessitates a data owner to pass on a filled in cipher-text section to each non-revoked user. They besides would not formulate a peculiarity flanked by personal and public domains.

2.2.1 Attribute Revocation

In recent times, numerous attributes revocable ABE formats have been planned [21], [15], [28]. They comprehend revocation by means of revoking attribute itself by means of timed rekeying method that is realized by

posting a cessation time on every attribute. This manuscript describes such a coarse-grained revocation for the reason that the instantaneous rekeying on any member modification might not be probable. To be sure, these techniques have two major predicaments. The primary dilemma is the security deprivation by means of forward and backward secrecy [29]. An attribute is supposed to be allocated in the midst of a group of users. Subsequently, it is a substantial set-up that membership might revolutionize recurrently in the faction that carves up an attribute. Afterward, a fresh consumer possibly will be competent to view the preceding data encrypted ahead of grasping the attributes awaiting the medical records are re-encrypted by means of freshly restructured attribute keys by sporadic rekeying. This would be termed as backward secrecy. Alternatively, a revoked user may possibly be capable of accessing the encrypted medical records albeit he would not embrace the attribute to further extent till the subsequent cessation time. This hysterical epoch is entitled as the window of vulnerability.

Scalability is yet another dilemma. The key authority sporadically proclaims a key update matter by means of unicast at each and every time slot with the intention that every non-revoked user can renew his key. This possibly will be a tailback for key authority and every non-revoked user. It is detected that such a scheme would be depreciated attributable to the fact that the preceding revocations were all set devoid of any contemplation of scalable circulation of renewed attribute keys towards collection of consumers who carve up the attributes. As a consequence, it remains an unresolved issue to devise a scalable as well as fine-grained revocation technique in the data outsourcing construction by means of ABE that is the primary predicament the MLPPT-MHS scheme would possibly endeavor to resolve in the anticipated work. Ibraimi et al. [13] along with Yu et al. [14] projected CP-ABE methodologies by way of instant attribute revocation competence more willingly than episodic or occasional revocation by means of the assistance of semi-trusted proxy positioned within the server. Conversely, they too have been unsuccessful in accomplishing fine-grained access control in outsourcing milieu.

2.2.2 User Revocation

In recent times, the noteworthiness of user revocation was perceived in numerous realistic methodologies. It is indeed an indispensable technique in myriad group-based application scenarios [29], [30] together with ABE systems, for the reason that users may alter their attributes recurrently. The fine-grained user revocation is accomplished by means of ABE that holds up unenthusiastic segments, anticipated in [31]. To finish that, one immediately appends the AND of the negation of annulled consumer identities. Conversely, this elucidation is moderately deficient in efficiency. Golle et al. [32] initiated a user revocable plot; however that plot barely fulfills the purpose when the amount of attributes equivalent to a cipher text is approximately partially the dimension of the universe. The preceding methods impose a restriction pertaining to the availability. Once a user is revoked as of an attribute group in the earlier designs [31], [24], he drops his access rights headed for the information sharing scheme. To be precise, the earlier techniques apprehended user revocation on top of system-level that would signify that as soon as a user is rescinded from an attribute group, he would be annulled from the intact organization. This setting is not so enviable like the attribute-level access control in scores of realistic outsourcing states of affairs, even though they comprehended immediate user revocation. Attrapadung and Imai [33] counseled a different user-revocable ABE scheme to get to the bottom of this dilemma by coalescing broadcast encryption and Attribute-based Encryption. Nevertheless, in this methodology, data possessor ought to acquire jam-packed accusation of preserving every association catalogs for every attribute group to facilitate user revocation. Such a method would never be pertinent to data outsourcing design, for the reason that the data holders would not unswervingly control the data distribution subsequent to outsourcing their data.

3. Cloud-based Information Sharing Framework

3.1 Problem Definition

The MLPPT-MHS scheme inspects a healthcare scheme which includes data owners as well as users. PHR owners pass onto doctors, who have chock-a-block authority on their patients' PHR details, i.e., they are capable of

generating, supervising, and erasing it. They own a vital server affiliated to the service provider that hoards every owner's PHRs. The consumers may perhaps approach from an assortment of features; for case in point, an acquaintance, a caregiver or an investigator. Patients access the PHR documents all the way in the course of the server so as to read/write to somebody's medical records; furthermore an owner can in chorus have access to compound data holders' medical records.

3.1.1 Security Mock-up

Within this manuscript, the authors mull over the server as honest-but-curious as in [34] and [17]. Server would endeavor to grasp a great deal of secret information in the health repository; nevertheless they may candidly tag along the protocol as a general rule. Alternatively, some consumers will as well endeavor to way in the files ahead of their privileges. For case in point, a pharmacy will perhaps like to accomplish the treatment plan of patients for marketing and enhancing its revenue. For that, they might conspire with supplementary users, or may be with the server.

3.1.2 Security Requirements

To pull off "patient-centric" PHR sharing, a central prerequisite is to facilitate individual patients to have power over who are allowed to view his/her individual PHR records. Principally, user-organized read/write access as well as revocation is the two major security aspirations intended for an electronic data system, designated by Mandl et al. [7] in 2001. The security as well as performance prerequisites are recapitulated as below:

- Data Confidentiality.** Illicit customers who fail to acquire satisfactory attributes gratifying the access policy or to encompass apposite key access license has to be thwarted from decrypting a medical record, in case of collusion. Access control must be put into effect, which implies that sundry consumers are endorsed to read unusual sets of records.
- Collusion Resistance:** Collusion resistance is yet another security asset entailed in ABE systems [11], [21], [41]. If manifold users conspire, they may know how to decrypt a cipher-text by mingling their attributes albeit the users are unable to decrypt the medical records on their own. The MLPPT-MHS scheme doesn't fancy colluders to know how to decrypt the confidential data by mingling these attributes.
- Forward and Backward Secrecy:** Forward secrecy connotes that whichever consumer who plunges an attribute is supposed to be banned from using the plaintext data handed out when he plunges the attribute except the supplementary applicable attributes that he is grasping gratify the access policy. Conversely, backward secrecy connotes that the consumer who approaches to grasp an attribute ought to be thwarted from using the plaintext of the earlier distributed data sooner than embracing the attribute.
- On-demand revocation.** When on earth a consumer's attribute is invalid, the consumer should not know how to view upcoming PHR files via that attribute. Such a scheme would be entitled attribute revocation; moreover the correspondent security asset is forward secrecy [25]. User revocation can also be performed, where the entire list of consumer's access rights would be annulled
- Write Access Control:** The MLPPT-MHS scheme will adjourn illegitimate donors to get write access to medical records, whilst the justifiable donors must contact the server by means of liability.
- Access Policies must be supple:** Vibrant modifications to the in-built strategies would be tolerable; principally the PHRs ought to be reachable under critical circumstances.

4. Preface

4.1 Cryptographic Backdrop

The MLPPT-MHS scheme initially endows with an official description of access structure by summarizing the descriptions in [11], [21]. Subsequently, the MLPPT-MHS scheme might momentarily appraise the cryptographic backdrop of the bilinear map in addition to security hypothesis.

4.1.1 Access Structure

Let us assume that $\{P_1, P_2, \dots, P_n\}$ denote a group of elements. A compilation $\mathcal{A} \subseteq 2^{\{P_1, P_2, \dots, P_n\}}$ would be monotone in the event that $\forall B, C$: allowing that $B \in \mathcal{A}$ as well as $B \subseteq C$ followed by $C \in \mathcal{A}$. Access structure would be a compilation $\mathcal{A}$ of nonempty subsets of $\{P_1, P_2, \dots, P_n\}$, to be precise, $\mathcal{A} \subseteq 2^{\{P_1, P_2, \dots, P_n\}} \setminus \{\emptyset\}$. Within the anticipated work, the responsibility of the elements is seized by means of attributes. Accordingly, the access structure $\mathcal{A}$ possibly would enclose the endorsed group of attributes.

4.1.2 Bilinear Pairings

Consider that G_1, G_2 , along with G_T denote three cyclic groups with prime order. Bilinear map is said to be map $e: G_1 * G_2 \rightarrow G_T$ that owns the subsequent characteristics.

1. *Bilinearity*: For all $g_1 \in G_1, g_2 \in G_2$, as well as $a, b \in \mathbb{Z}_p^*$, $e(g_1^a, g_2^b) = e(g_1, g_2)^{ab}$.
2. *Non-degeneracy*: $e(g_1, g_2) \neq 1$.
3. *Computability*: There subsists a proficient scheme to work out $e(g_1, g_2)$ in favor of any $g_1 \in G_1$ and $g_2 \in G_2$.

4.1.3 Bilinear Diffie-Hellman Hypothesis

By means of the mentioned entries, Bilinear Diffie-Hellman dilemma is supposed to work out $e(g, g)^{abc} \in G_T$ specified a generator of G along with the entities g^a, g^b, g^c for $a, b, c \in \mathbb{Z}_p^*$. A counterpart formation of the Bilinear Diffie Hellman predicament would be to figure $e(A, B)^c$ known a generator of G , as well as entities A, B and g^c in G . Algorithm $\mathcal{A}$ possesses an advantage $\epsilon(\kappa)$ in deciphering Bilinear Diffie Hellman dilemma in favor of a bilinear map group $\langle p, G, G_T, e \rangle$, in which κ is the security restriction, if $\Pr[\mathcal{A}(p, G, G_T, A, B, g^c) = e(A, B)^c] \geq \epsilon(\kappa)$. On the assumption that for each polynomial-time algorithm to resolve the Bilinear Diffie Hellman dilemma on $\langle p, G, G_T, e \rangle$, advantage $\epsilon(\kappa)$ would be a trifling function, subsequently $\langle p, G, G_T, e \rangle$ is alleged to gratify Bilinear Diffie Hellman postulation.

4.2 Preliminaries

Let us assume that $\mathcal{U} = \{u_1, \dots, u_n\}$ denote cosmos of users. Let us assume that $L = \{\lambda_1, \dots, \lambda_p\}$ denote space of expressive attributes prevalent in the organization. Let us assume that $G_i \subset \mathcal{U}$ denote a group of consumers who embrace the attribute λ_i that is delineated as being an attribute group. G_i would be employed as an access list otherwise revocation list to λ_i . Entitle $\mathcal{G} = \{G_1, \dots, G_p\}$ as the creation of attribute groups. Entitle K_{λ_i} as the attribute group key that would be allocated in the midst of the non-revoked users in $G_i \in \mathcal{G}$.

4.2.1 Cipher-text Policy Attribute-Based Encryption

The CP-ABE proposal which is characterized below is composed of the subsequent deep-seated phases.

1. **Setup:** Setup phase receives no input excluding the inherent security entity. This yields public parameters PK along with a master key which is denoted by MK .
2. **Encrypt ($PK, M, \mathcal{A}$):** This phase receives as load public parameters, message, along with access structure. This would encrypt the message as well as spawn a medical record so that merely a consumer that grasps a group of attributes that kowtows to access structure would know how to decrypt the communication. It would be presumed that the cipher-text unreservedly encloses $\mathcal{A}$.
3. **Key Generation ($MK, \mathcal{S}$):** The third phase receives as load master key along with a group of attributes to depict the key. This yields a private key of length 16/24/32 bits every time. A randomness function is engaged to attain this varied key length.
4. **Decrypt (PK, CT, SK):** Decryption phase receives public parameters, cipher-text that restrains an access policy, along with a private key in favor of a group $\mathcal{S}$ of attributes as load. In the event that group of attributes accomplishes the access structure, this phase can decrypt the cipher-text and endow with a message M .
5. **Delegate ($SK, \mathcal{S}$):** Delegate phase receives a secret key SK on behalf of a little group of attributes along with a set $\mathcal{S} \subseteq S$ as load. This yields a secret key in favor of the group of attributes $\mathcal{S}$.

4.2.2 Security Model

Setup: Challenger performs Setup phase as well as endows with the public parameters to the antagonist.

Phase I: Antagonist crafts recurring private keys analogous to the set of attributes $S_1, \dots, S_{q1}$.

Challenge: Antagonist tenders two identical span messages M_0 as well as M_1 . Into the bargain, the antagonist proffers a challenge access structure $\mathcal{A}^*$ so that not an iota of $S_1, \dots, S_{q1}$ gratify the access structure. Challenger turns over a capricious coin b as well as encrypts M_b beneath $\mathcal{A}^*$. The cipher-text CT^* would be furnished towards the antagonist.

Phase II: The phase I is reiterated by way of the constraint that not an iota of the assortment of attributes $S_{q1+1}, \dots, S_q$ assures access structure equivalent to the challenge.

Guess: The antagonist yields a guess b' of b .

The benefit of an antagonist $\mathcal{A}$ is characterized as $\Pr[b' = b] - \frac{1}{2}$. It is observed that the replica would effortlessly be pulled out on the way to hold chosen-cipher-text hits by permitting for decryption inquiries in Phase I as well as Phase II.

Definition: A CP-ABE technique is safe and sound if each and every polynomial time antagonists have at most a trifling advantage in the above mentioned game.

5. Overview of MLPPT-MHS

In this fragment, the authors endow with the edifice of our format to some extent anchored in nevertheless not restricted to Yang et al.'s edifice [36] so as to boost the eloquence of access structure. The anticipated plot espouses a twofold encryption methodology to surmount access control predicament.

5.1 Access Tree

5.1.1 Portrayal

Assume that T denotes a tree indicating an access structure. Every non-leaf node embodies a threshold gate. In the event that num_x is the quantity of children in node x along with k_x as threshold value, in that case $0 \leq k_x \leq num_x$. Every leaf node is portrayed by means of attribute in addition to a porch detail $k_x = 1$. λ_x signifies attribute allied amid the leaf node. $p(x)$ embodies its parent. Its children are denoted as 1 to num . $index(x)$ yields that value coupled with entity x . Index details are inimitably dispensed to entities in access structure in favor of a prearranged key in a capricious conduct.

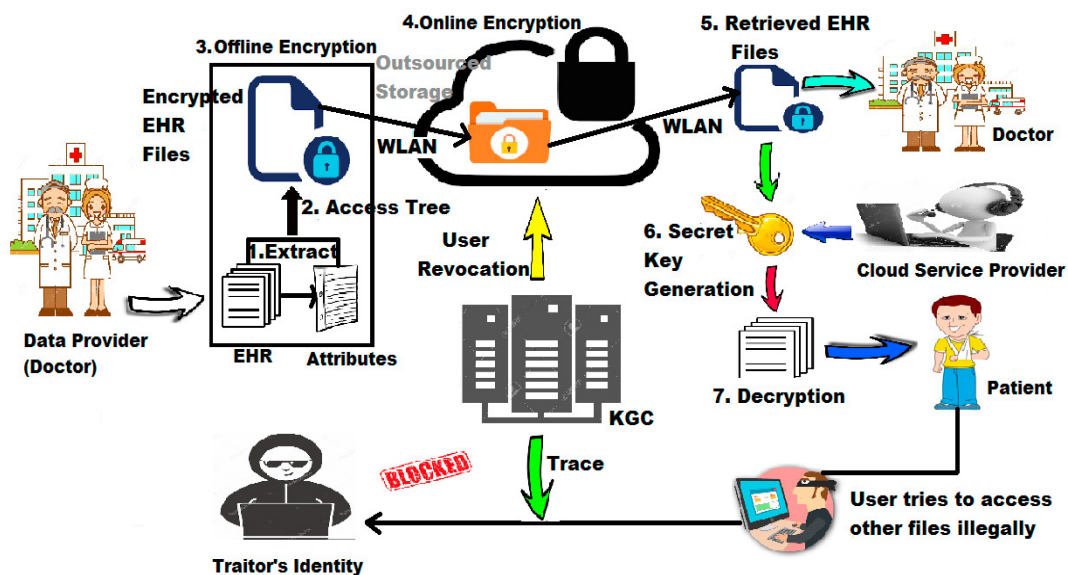


Figure 1: MLPPT-MHS System Architecture

5.1.2 Gratifying an Access Tree

Let T_x denote sub-tree of T mounted by the side of the entity x . In the event that a assemblage of attributes Y persuades T_x , it is symbolized as $T_x(Y) = 1$. $T_x(Y)$ is calculated recurrently along these lines: In the event that x is a non-leaf node, appraise $T_x(Y)$ in favor of every single children x' of node x . $T_x(Y)$ yields 1 iff at least k_x children yield 1. In the event that x acts as a leaf node, in that case $T_x(Y)$ yields 1 iff $\lambda_x \in Y$.

5.2 Design Edifice

Let G denote a bilinear group consisting of a prime order p ; furthermore assume that g denote the generator of G . Entitle $e: G * G \rightarrow G_T$ as the bilinear map. The security entity, κ , would terminate the magnitude of the groups. They would furthermore take advantage of Lagrange coefficients $\Delta_{i,\wedge}$ in favor of any $i \in Z_p^*$ along with a list $\wedge$ of entities in Z_p^* ; delineate $\Delta_{i,\wedge}(x) = \prod_{j \in \wedge, j \neq i} (x-i)/(i-j)$. This would as well provide work for a hash function $H: \{0, 1\}^* \rightarrow G$ to bracket together each and every attribute with an arbitrary group entity in G that would be sculpted like a random

oracle.

5.2.1 System Layout

All through the primary layout phase, trusted authority executes the Setup phase. It picks a bilinear group with order in the midst of generator g . Afterward, it picks random exponents $\alpha, \beta \in \mathbb{Z}_p^*$. Public parameters include $G, g, h = g^\beta, e(g, g)^\alpha$ along with master key is (β, g^α) . Though the public parameter is left out for key delegation, it would be effortlessly pulled out to accomplish the workability of key delegation by issuing additional supplementary entity; e.g., $g^{1/\beta}$ in [21].

5.2.2 Key Generation

This segment comprises attribute key generation via the Key Generation Center (KGC). Subsequent to creation of the system public as well as secret parameters, trust authority engenders attribute keys on behalf of a group of consumers by executing $\text{KeyGen}(\text{MK}, \wedge, U)$ phase. This receives a group of attributes $\wedge \subseteq \mathcal{U}$ as load; furthermore it yields an attribute key for every consumer that recognizes with the corresponding set $\wedge$. It initially prefers an arbitrary $r \in \mathbb{Z}_p^*$ (that is inimitable to individual user), in addition to an indiscriminate $r_j \in \mathbb{Z}_p^*$ for each and every attribute $\lambda_j \in \wedge$. After that, this provides the subsequent secret value to the user $u_t \in U$:

$$SKt = (D = g(\alpha+r)/\beta, \forall \lambda_j \in \wedge : Dj = g^r \cdot H(\lambda_j)^{r_j}, D'_j = g^{r_j}).$$

The trust authority provides attribute groups G_j for every $\lambda_j \in \wedge$ to the data service manager. For case in point, u_1, u_2, u_3 would be allied among $\{\lambda_1, \lambda_2, \lambda_3\}, \{\lambda_2, \lambda_3\}, \{\lambda_1, \lambda_3\}$, correspondingly, the trusted authority provides $G1 = \{u_1, u_3\}, G2 = \{u_1, u_2\}$, along with $G3 = \{u_1, u_2, u_3\}$ to the data service manager.

5.2.3 Outsourcing the Data

As soon as a data holder desires to contract out its data, he characterizes tree access structure beneath the cosmos of attributes as well as encrypts the medical records by executing $\text{Encrypt}(\text{PK}, M, \mathcal{T})$ phase. This phase implements access control by means of speckled key length on the outsourced data.

5.2.4 Re-Encrypting the Data

Prior to doling out the contracted data, the service manager would re-encrypt the contracted out data online by means of a group of the membership details apiece $G \subseteq \mathcal{G}$ that materializes within the access tree entrenched within the cipher text. The scheme advances in the following manner. The cipher-text attained from the offline phase is granted as input to the online phase. The encrypted data CT in transit is sent through a secure tunnel encrypted through SSL/TLS and the encrypted data CT at rest is encrypted by means of Advanced Encryption Standard 256-bit (AES-256).

5.2.5 Data Decryption

The decryption process is segregated into two segments: offline as well as online. In the online segment, the data

is decrypted using the AES-256 algorithm. The offline decryption happens in the following manner. The decryption phase obtains as load the public entities, cipher-text that encloses access policy, in addition to private key. In the event that the group S of attributes convinces the access structure after that this phase will decrypt the cipher-text as well as yield a message M .

6. Security Dissection

This fragment would thrash out the security concerning the anticipated method pertaining to security prerequisites argued in Section II.

6.1 Collusion Resistance

Akin to the preceding ABE methodologies [21], [41], the private keys of consumers are randomized by the Key Generation Center so that those would not be coalesced in the anticipated plan. With the intention of decrypting a cipher-text, the colluding attackers ought to decrypt the concealed data by merging their attributes. In view of the fact that the Key Generation Center (KGC) is honest, the MLPPT-MHS scheme would never deem any active attacks through conspiring with revoked users.

6.2 Data Confidentiality

The solitary procedure for imposing that policy is to occupy a trusted server to stock up medical records as well as arbitrate access control. Conversely, in the event that any server holding the medical record is accommodated, the data confidentiality would be compromised. Data confidentiality against outsiders who never have an adequate amount of attributes can be immaterially guaranteed.

6.3 Forward and Backward Secrecy

Albeit the consumer has amassed the preceding cipher-text earlier, he gets hold of the attribute keys along with attributes assure the access policy, he is not capable of decrypting the preceding cipher-text. This is for the reason that, even though he can accomplish something in computing from the contemporary cipher-text, it would not lend a hand to recuperate the preferred detail for the prior cipher-text in view of the fact that it is deprived via a haphazard. Consequently, the backward secrecy is surefired in the anticipated method. In contrast, as soon as a user moves toward to plunge a group of attributes convincing the access policy at some case in point, equivalent attribute group keys are too rehabilitated as well as disseminated to the legitimate attribute entities steadily. Afterward, the consumer would not be able to decrypt any data parallel to the attributes subsequent to his revocation attributable to the loss of sight affected from freshly updated attribute group keys.

6.4 On-demand Revocation

On every occasion a user's attribute becomes invalid, the user is not supposed to know how to view further health data via the corresponding attribute. This would be characteristically termed attribute revocation; moreover the analogous security asset would be forward secrecy [25]. There exists user revocation, in which access privileges of the entire users are revoked.

6.5 Write Access Control

The MLPPT-MHS scheme would avert the illegal contributors to expand write-access to data holders' PHRs, at the same time as the justifiable suppliers must access the server in the midst of accountability.

6.6 Supple Access Policy

Vibrant modifications to the in-built policies would be tolerated; above it is mandatory for all the PHRs to be accessible under tragedy.

7. Scheme Analysis

In this segment, the MLPPT-MHS scheme is investigated as well as judged against the functionalities of the preceding CP-ABE methodologies. The MLPPT-MHS scheme furthermore talks about its effectiveness once employed in the midst of precise factors as well as weigh against these results in the midst of those acquired by the supplementary schemes.

Table 1. Function Comparison.

Scheme	Access Control	User Specific Attributes	Random Key Length	Trace	User Revoke
ABEN [31]	✓	✗	✗	✗	✗
IPSD [47]	✓	✗	✗	✗	✗
ODAC [48]	✓	✗	✗	✗	✗
AVOD [49]	✓	✗	✗	✗	✗
AEVD [50]	✓	✗	✗	✗	✗
GEAV [51]	✓	✗	✗	✗	✗
WTCM [43]	✓	✗	✗	✓	✗
WTCF [44]	✓	✗	✗	✓	✗
[46]	✓	✗	✗	✗	✗
VAKF [45]	✓	✗	✗	✗	✓
LSTM [36]	✓	✗	✗	✓	✓
MLPPT-MHS	✓	✓	✓	✓	✓

7.1 Key Escrow with Revocation

Key Generation Centre possibly will decrypt any data destined to unambiguous consumers via engendering the corresponding private keys which would be recognized as the key escrow problem. Another challenge is the key revocation. This would not be appropriate in data sharing circumstances wherein the data owner would be fond of composing their private data only accessible to nominated consumers. Within the anticipated scheme, the behaviour of the KGC is monitored through the system to reduce the illegal behaviour of KGC.

Table 2. Storage Overhead Comparison.

Scheme	PP	SK	CT
ABEN [31]	$(2 \mathcal{U} + 3) G $	$(5 S) G $	$(2l+1) G + G_T $
IPSD [47]	$(2 \mathcal{U} + 4) G + G_T $	$(S + 6) G $	$(5l) G + G_T $
ODAC [48]	$2 G + G_T $	$(S + 2) G $	$(2l+1) G + G_T $
AVOD [49]	$(\mathcal{U} + 5) G + G_T $	$(S + 2) G $	$(4l+3) G + 2 G_T $
AEVD [50]	$(\mathcal{U} + 2) G + G_T $	$(S + 2) G $	$(2l+1) G + G_T $
GEAV [51]	$(\mathcal{U} + 4) G + G_T $	$(S + 2) G $	$(2l+1) G + G_T $
WTCM [43]	$(\mathcal{U} + 3) G + G_T $	$(S + 3) G + Z_p $	$(2l+2) G + G_T $
WTCF [44]	$6 G + G_T $	$(2 S + 3) G + Z_p $	$(3l+2) G + G_T $

SADS [46]	$(2 \mathcal{U} +10) G + 3 G_T $	$(3 S) G $	$(l+3) G + 2 G_T $
VAKF [45]	$(3 \mathcal{U} +1) G + G_T $	$(2 S +1) G + Z_p $	$(l+2) G $
LSTM [36]	$3 G + 2 G_T $	$(S +1) G + 7 Z_p $	$2 G + 2 G_T + 2 Z_p $
MLPPT-MHS	$2 G + G_T $	$(3 S) G + Z_p $	$2 G + G_T + Z_p $

Table 3. Computation Overhead Comparison.

Scheme	KeyGen	Encryption	Decryption
ABEN [31]	$(6 S)t_{el}$	$t_p + t_{e2} + (2l+1)t_{el}$	$(3 S)t_p + 2 S t_{e2} + S t_{el}$
IPSD [47]	$(S +9)t_{el}$	$t_p + lt_{e2} + (6l)t_{el}$	$(6 S)t_p + 2 S t_{e2}$
ODAC [48]	$(S +2)t_{el}$	$t_p + t_{e2} + (3l+1)t_{el}$	t_{el}
AVOD [49]	$(S +3)t_{el}$	$2t_p + 2t_{e2} + (6l+4)t_{el}$	t_{el}
AEVD [50]	$(S +3)t_{el}$	$t_p + t_{e2} + (3l+1)t_{el}$	t_{el}
GEAV [51]	$(S +3)t_{el}$	$t_p + t_{e2} + (3l+3)t_{el}$	t_{el}
WTCM [43]	$(S +4)t_{el}$	$t_p + t_{e2} + (3l+2)t_{el}$	$(2 S +l)t_p + S t_{e2} + (S +1)t_{el}$
WTCF [44]	$(4 S +4)t_{el}$	$t_p + t_{e2} + (5l+2)t_{el}$	$(3 S +l)t_p + S t_{e2} + (S +1)t_{el}$
SADS [46]	$4 S t_{el}$	$2t_p + 2t_{e2} + (l+5)t_{el}$	$4t_p + t_{e2} + (3l+4)t_{el}$
VAKF [45]	$(2 S +2)t_{el}$	$(2 S +2)t_{el}$	$\perp$
LSTM [36]	$(S +1)t_{el}$	$3t_{e2} + 2t_{el}$	t_{el}
MLPPT-MHS	$(3 S)t_{el}$	$2t_{e2} + t_{el}$	t_{el}

7.2 Comparison

Tables 1-3 judge against the functionality, computation and storage overhead of MLPPT-MHS with other formats that would put forth access control over the concealed data.

Table 4. Storage Overhead (bits) ($|S| = 100$).

Scheme	WTCM [43]	WTCF [44]	SADS [46]	VAKF [45]	LSTM [36]	MLPPT-MHS
$ PP $	106,496	8,192	218,112	309,248	7,168	6,098
$ SK $	105,632	208,032	307,200	207,008	104,544	59,653
$ CT $	207,008	309,408	105,792	104,448	36,096	26,712

The notations used in Table 1-3 are defined below. Let $|PP|$, $|SK|$, $|CT|$ indicate the magnitudes of public parameter, secret key of user, and the cipher-text correspondingly. Let us assume that l represent the amount of rows in the matrix of the access structure, $|\mathcal{U}|$ denote dimension of U , moreover $|S|$ denote size of attribute set S . $|G|$, $|G_T|$ along with $|Z_p|$ symbolizes the bit length of an entity in group G , G_T , along with Z_p , correspondingly. Represent t_{e1} , t_{e2} and t_p as modular exponentiation on G and G_T along with a bilinear pairing, correspondingly.

Table 5. Computation Time (ms) ($|S| = 100$).

Scheme	WTCM [43]	WTCF [44]	SADS [46]	VAKF [45]	LSTM [36]	MLPPT-MHS
<i>KeyGen</i>	1,118	1,055	4,093	2,072	906.67	823.75
<i>Enc</i>	27,444	45,377	9,919	18,203	280.43	152.43
<i>Dec</i>	51,657	71,167	28,209	$\perp$	90.11	82.54

Table 2 shows the storage overhead comparison. It can be seen that MLPPT-MHS has less significant public parameter size, secret key size, in addition cipher-text size. A detailed analysis is mentioned below.

1. **Public Parameter Size:** It can be seen that [48], [44], [36], and MLPPT-MHS supports a boundless amount of attributes in the scheme. Such an aspect would in fact be obliging for the huge-scale mHealth network in view of the fact that the public parameter size is incontrovertible in the midst of the size of attribute set.

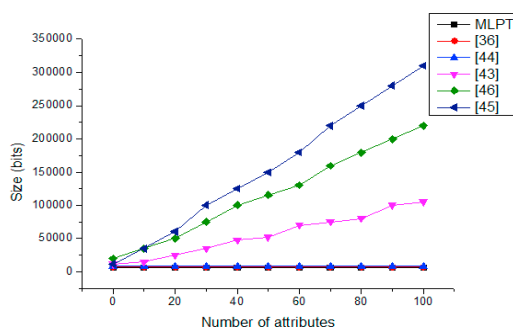


Figure 2: Storage and Transmission Overhead

- Public Parameter Size

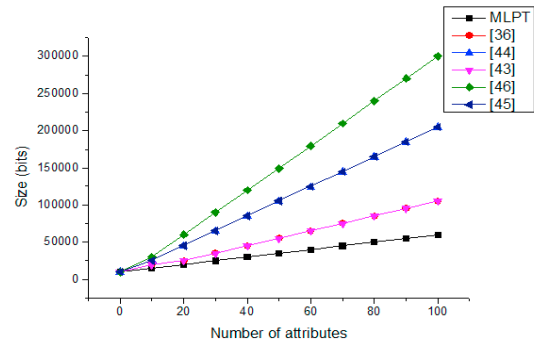


Figure 3: Storage and Transmission Overhead

- Secret Key Size

2. **Secret Key Size:** A user's secret key embraces $(3|S|)$ entities in group G along with one entity in Z_p . By and large, $|Z_p|$ is constantly smaller than $|G|$, which will be investigated further below. It is apparent that the magnitude of the consumer's secret key in MLPPT-MHS is lesser in contrast to the schemes in [44] – [46]. Less significant secret key size as well connotes lesser storage overhead in mobile devices.

3. **Cipher-text Size:** The cipher-text spawned in MLPPT-MHS has two entities in group G , one entity in group G_T along with one entity in Z_p . Since the $|Z_p|$ is characteristically as a minimum one-sixth of $|G|$, $|CT|$ in MLPPT-MHS is tinier when compared to the previous methods within Table 2. Consequently, MLPPT-MHS necessitates little storage overhead and minor transmission cost flanked by the data owner as well as the public cloud.

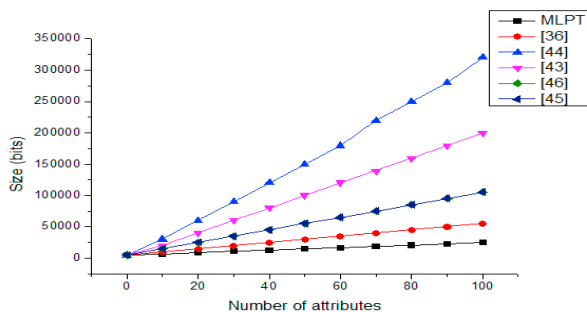


Figure 4: Storage and Transmission Overhead

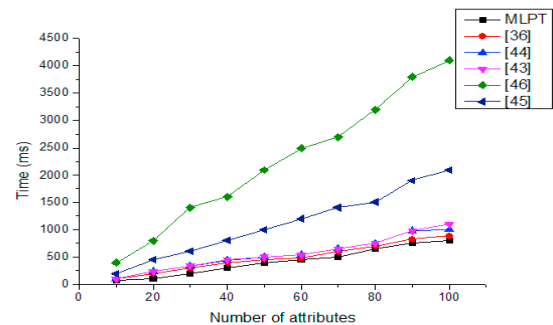


Figure 5: Computation Overhead

- Cipher Text Size

- Key Generation Time

As illustrated in Table 3, MLPPT-MHS has lesser computation overhead in each algorithm in contrast to other methods.

- **KeyGen:** The KGC possibly will make the most of $(3|S|)$ exponentiation operations on G to engender secret key. Every single supplementary design in Table 2 entail supplementary computation than ours. For case in point, [44], [46] requires as much as $4|S| + 4$ along with $4|S|$ exponentiations on G during the *KeyGen* phase, correspondingly.
- **Encryption:** Within MLPPT-MHS, encryption of health data is prepared by means of the user's energy restricted node. In MLPPT-MHS, no pairing computation is engrossed during the encryption phase. Besides, no more than one exponentiation on G along with two exponentiations on G_T are obliged to produce a cipher-text.
- **Decryption:** Consuming the outsourced decryption mechanism, the methods in [48] – [51], [36], and MLPPT-MHS can recuperate the EHR with a single exponentiation calculation on group G .

7.3 Experimental Analysis

Experimentations are carried out on PC and smartphone to estimate the performance of MLPPT-MHS. PC is exploited to conjure up the public cloud as well as KGC, which process comparatively elevated computation competence in addition to eternal electricity supply. The smartphone is the mobile agent of data owner or consumer with minimal computation constraints as well as restricted battery.

7.3.1 Experimental Results

- Storage and Transmission Competence:** Headed for assessing the storage as well as transmission overhead, MLPPT-MHS is matched up with the schemes [43] – [46] by means of public parameter size, secret key size, as well as cipher-text size (bits) in Figure 2 as well as Table 4. With the intention of portraying its execution, a non-uniform axis is employed within Figure 2 in the direction of formulating the dissimilar values way better.

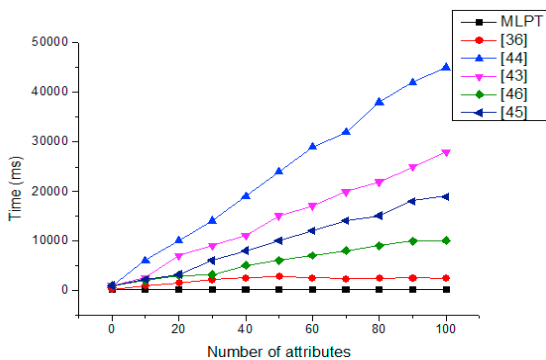


Figure 6: Computation Overhead

- Encryption Time

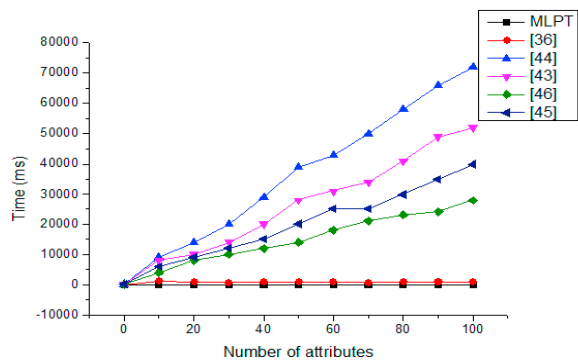


Figure 7: Computation Overhead

- Decryption Time

Figure 2 demonstrates that regardless of the number of attributes, the public parameter size is 6098 bits. Figure 3 comes across like there are merely four streaks for six methods. This is because the schemes in [44], [45] have analogous size of SK along with these two streaks lie on top of each other. Once $|S| = 100$, [46] necessitates 307,200 bits for storing secret key that is five times of our scheme's secret key storage. Figure 4 designates that MLPPT-MHS necessitates smallest amount of storage space for cipher-text that is functional to hoard funds in the pay-per-use cloud.

b. Computational Efficiency:

As illustrated in Figure 5-7, the anticipated scheme has implemented each and every algorithm in MLPPT-MHS and that in [36], [43] - [46]. The experimentations on PC as well as Smartphone signify that similar fundamental computations (like bilinear pairing along with exponentiation) performed by PC is about ten times more rapid when compared to that on Smartphone. The experimental results are illustrated in Figure 5-7 and Table 5 which undoubtedly exhibits that MLPPT-MHS constantly incurs slightest computation time in contrast to others.

- As signified in Table 5, encryption time of the proposed scheme is about 152.43 ms furthermore it does not fluctuate in the midst of the count of attributes. Conversely, encryption duration devours 27,444 ms, 45,377 ms, 9,919 ms, 18,203 ms, along with 280.43 ms in [43], [44], [46], [45], and [36], correspondingly.
- In the *Decryption* algorithm, MLPPT-MHS has a steady computation cost 82.54 ms to perform decryption. Conversely, the techniques in [43], [44], [46] calls for 51,657 ms, 71,167 ms, as well as 28,209 ms to convalesce a single medical record, correspondingly. The huge calibration overhead would rapidly consume the battery of the consumer's resource-limited mobile gadget.

This investigation explains that MLPPT-MHS has efficiency significantly superior than its preceding schemes.

Conclusion

The imposition of access policies in addition to the assistance of policy revisions are imperative exigent concerns in a typical record sharing system. Within this manuscript, a multi-layered privacy-preserving and traceable mobile health system is anticipated to insist on a fine-grained access control by taking advantage of the traits of any data sharing system. Bearing in mind that mobile gadgets in healthcare applications are resource restricted, maneuvers in holders' and consumers' devices in MLPPT-MHS are set aside as frivolous. The anticipated method can perform an instantaneous consumer revocation on each and every attribute group whilst taking jam-packed benefit of the scalable access control. Then the official characterization of the security of MLPPT-MHS as well as confirmation of its security is verified devoid of random oracle. The subjective investigation illustrated that MLPPT-MHS is surpassing the majority of the preceding techniques. Far-reaching research on its routine (on PC as well as mobile) revealed that MLPPT-MHS shows potential for sensible applications.

References

- [1] M. Li, S. Yu, K. Ren, and W. Lou. (2010) "Securing Personal Health Records in Cloud Computing: Patient-Centric and Fine-Grained Data Access Control in Multi-Owner Settings." *Proceedings of the Sixth International ICST Conference on Security and Privacy in Comm. Networks* 89-106.
- [2] H. Lohr, A.-R. Sadeghi, and M. Winandy. (2010) "Securing the E-Health Cloud." *Proceedings of the First ACM International Health Informatics Symposium* 220-229.
- [3] M. Li, S. Yu, N. Cao, and W. Lou. (2011) "Authorized Private Keyword Search over Encrypted Personal Health Records in Cloud Computing." *Proceedings of the 31st International Conference of Distributed Computing Systems*.
- [4] "The Health Insurance Portability and Accountability Act," http://www.cms.hhs.gov/HIPAAGenInfo/01_Overview.asp, 2012
- [5] "Google, Microsoft Say HIPAA Stimulus Rule Doesn't Apply to Them," <http://www.ihealthbeat.org/Articles/2009/4/8/>, 2012.
- [6] "At Risk of Exposure - in the Push for Electronic Medical Records, Concern Is Growing about How Well Privacy Can Be Safeguarded," <http://articles.latimes.com/2006/jun/26/health/he-privacy26>, 2006.
- [7] K.D. Mandl, P. Szolovits, and I.S. Kohane. (2001) "Public Standards and Patients' Control: How to Keep Electronic Medical Records Accessible but Private." *BMJ* **322(7281)**: 283-287.

- [8] J. Benaloh, M. Chase, E. Horvitz, and K. Lauter. (2009) "Patient Controlled Encryption: Ensuring Privacy of Electronic Medical Records." *Proceedings of the ACM Workshop Cloud Computing Security* 103-114.
- [9] S. Yu, C. Wang, K. Ren, and W. Lou. (2010) "Achieving Secure, Scalable, and Fine-grained Data Access Control in Cloud Computing." *Proceedings of the IEEE INFOCOM '10*.
- [10] C. Dong, G. Russello, and N. Dulay. (2010) "Shared and Searchable Encrypted Data for Untrusted Servers." *Journal of Computer Security*, **19**(1):367-397.
- [11] V. Goyal, O. Pandey, A. Sahai, and B. Waters. (2010) "Attribute-Based Encryption for Fine-Grained Access Control of Encrypted Data." *Proceedings of the 13th ACM Conference on Computer and Communications Security* 89-98.
- [12] M. Li, W. Lou, and K. Ren, "Data Security and Privacy in Wireless Body Area Networks." *IEEE Wireless Communication Magazine*, **17**(1):51-58.
- [13] L. Ibraimi, M. Petkovic, S. Nikova, P. Hartel, and W. Jonker. (2009) "Mediated Ciphertext-Policy Attribute-Based Encryption and Its Application." *Proceedings of the International Workshop Information Security Applications* 309-323.
- [14] S. Yu, C. Wang, K. Ren, and W. Lou. (2010) "Attribute-Based Data Sharing with Attribute Revocation." *Proceedings of the ACM Symposium on Information, Computer, and Communication Security*.
- [15] A. Boldyreva, V. Goyal, and V. Kumar. (2008) "Identity-Based Encryption with Efficient Revocation." *Proceedings of the 15th ACM Conference on Computer and Communication Security* 417-426.
- [16] L. Ibraimi, M. Petkovic, S. Nikova, P. Hartel, and W. Jonker. (2009) "Ciphertext-Policy Attribute-Based Threshold Decryption with Flexible Delegation and Revocation of User Attributes."
- [17] S. Yu, C. Wang, K. Ren, and W. Lou. (2010) "Attribute-Based Data Sharing with Attribute Revocation." *Proceedings of the Fifth ACM Symp. Information, Computer, and Communication Security*.
- [18] S. Narayan, M. Gagne', and R. Safavi-Naini. (2010) "Privacy-Preserving EHR System Using Attribute-Based Infrastructure." *Proceedings of the ACM Cloud Computing Security Workshop* 47-52.
- [19] X. Liang, R. Lu, X. Lin, and X.S. Shen. (2010) "Patient Self-controllable Access Policy on Phi in E-healthcare Systems." *Proceedings of the Advances in Health Informatics Conference*.
- [20] L. Ibraimi, M. Asim, and M. Petkovic. (2009) "Secure Management of Personal Health Records by Applying Attribute-Based Encryption." *Technical Report*, University of Twente.
- [21] J. Bethencourt, A. Sahai, and B. Waters. (2007) "Ciphertext-Policy Attribute-Based Encryption," *Proceedings of the IEEE Symposium on Security and Privacy* 321-334.
- [22] J.A. Akinyele, C.U. Lehmann, M.D. Green, M.W. Pagano, Z.N.J. Peterson, and A.D. Rubin. (2010) "Self-Protecting Electronic Medical Records Using Attribute-Based Encryption." *Cryptology ePrint Archive*, Report 2010/565, <http://eprint.iacr.org/>
- [23] M. Chase and S.S. Chow. (2009) "Improving Privacy and Security in Multi-Authority Attribute-Based Encryption." *Proceedings of the 16th ACM Conference on Computer and Comm. Security*:121-130.
- [24] X. Liang, R. Lu, X. Lin, and X.S. Shen. (2010) "Ciphertext-Policy Attribute-Based Encryption with Efficient Revocation." *Technical Report, University of Waterloo*, <http://bbcr.uwaterloo.ca/~x27liang/papers/abe/%20with%20revocation.pdf>
- [25] J. Hur and D.K. Noh. (2011) "Attribute-Based Access Control with Efficient Revocation in Data Outsourcing Systems." *IEEE Transactions on Parallel and Distributed Systems*, **22**(7):1214-1221.
- [26] S. Jahid, P. Mittal, and N. Borisov. (2011) "Easier: Encryption-Based Access Control in Social Networks with Efficient Revocation." *Proceedings of the ACM Symposium on Information, Computer, and Communication Security*.
- [27] S. Ruj, A. Nayak, and I. Stojmenovic. (2011) "DACC: Distributed Access Control in Clouds." *Proceedings of the IEEE 10th International Conference on Trust, Security, and Privacy in Computing and Communication*.
- [28] M. Pirretti, P. Traynor, P. McDaniel, and B. Waters. (2006) "Secure Attribute-Based Systems." *Proceedings of the ACM Conference on Computer and Communication Security*.
- [29] S. Rafaeli and D. Hutchison. (2003) "A Survey of Key Management for Secure Group Communication." *ACM Computing Surveys*, **35**(3):309-329.
- [30] D. Naor, M. Naor, and J. Lotspiech. (2001) "Revocation and Tracing Schemes for Stateless Receivers." *Proceedings of the International Cryptology Conference on Advances in Cryptology* 41-62.
- [31] R. Ostrovsky, A. Sahai, and B. Waters. (2007) "Attribute-Based Encryption with Non-Monotonic Access Structures." *Proceedings of the ACM Conference on Computer and Communication Security* 195-203.
- [32] P. Golle, J. Staddon, M. Gagne, and P. Rasmussen. (2008) "A Content-driven Access Control System." *Proceedings of the Symposium on Identity and Trust on the Internet* 26-35.
- [33] N. Attrapadung and H. Imai. (2009) "Conjunctive Broadcast and Attribute-Based Encryption." *Proceedings of the International Conference Palo Alto on Pairing-Based Cryptography* 248-265.
- [34] S.D.C. di Vimercati, S. Foresti, S. Jajodia, S. Paraboschi, and P. Samarati. (2007) "Over-Encryption: Management of Access Control Evolution on Outsourced Data." *Proceedings of the 33rd International Conference on Very Large Data Bases* 123-134.
- [35] "Indivo." <http://indivohealth.org/>, 2012.
- [36] Yang Yang, Ximeng Liu, Robert H. Deng, and Yingjiu Li. (2017) "Lightweight Sharable and Traceable Secure Mobile Health System" *IEEE Transactions on Dependable and Secure Computing*.
- [37] A. Shamir. (1984). "Identity-Based Cryptosystems and Signature Schemes." *Proceedings of the Advances in Cryptology* **196**:37–53.
- [38] D. Boneh and M. Franklin. (2001) "Identity-Based Encryption from the Weil Pairing." *Proceedings of the Advances in Cryptology* **139**:213–229.

- [39] C. Cocks. (2001) “An Identity-based Encryption Scheme based on Quadratic Residues,” *Proceedings of the IMA International Conference* 360–363.
- [40] K.C. Almeroth and M.H. Ammar. (1997) “Multicast Group Behavior in the Internet’s Multicast Backbone (MBone),” *IEEE Communications Magazine* **35**(6):124-129.
- [41] A. Sahai and B. Waters. (2005) “Fuzzy Identity-Based Encryption.” *Proceedings of the International Conference Theory and Applications of Cryptographic Techniques* 457-473.
- [42] S. Yu, C. Wang, K. Ren, and W. Lou. (2010) “Attribute-Based Data Sharing with Attribute Revocation.” *Proceedings of the ACM Symposium on Information, Computer, and Communications Security*.
- [43] Z.Liu, Z. Cao, D. Wong. (2013) “White-box Traceable Ciphertext-policy Attribute-based Encryption Supporting Any Monotone Access Structures.” *IEEE Transactions on Information Forensics and Security* **8**(1): 76-88.
- [44] J. Ning, X. Dong, Z. Cao, L. Wei. (2015) “White-box Traceable Cipher-text Policy Attribute-based Encryption Supporting Flexible Attributes.” *IEEE Transactions on Information Forensics and Security* **10**(6):1274-1288.
- [45] W. Sun, S. Yu, W. Lou, Y. Hou, and H. Li. (2016) “Protecting Your Right: Verifiable Attribute-based Keyword Search with Fine-grained Owner-enforced Search Authorization in the Cloud.” *IEEE Transactions on Parallel and Distributed Systems* **27**(4):1187-1198.
- [46] K. Liang, W. Susilo. (2015) “Searchable Attribute-Based Mechanism with Efficient Data Sharing for Secure Cloud Storage.” *IEEE Transactions on Information Forensics and Security* **10**(9):1981-1992.
- [47] J. Han, W. Susilo, Y.Mu. (2015) “Improving Privacy and Security in Decentralized Ciphertext-Policy Attribute-based Encryption.” *IEEE Transactions on Information Forensics and Security* **10**(3): 665-678.
- [48] M. Green, S. Hohenberger, B. Waters. (2011) “Outsourcing the Decryption of ABE Ciphertexts.” *Proceedings of the USENIX Security Symposium*.
- [49] J. Lai, R. H. Deng, C. Guan, J. Weng. (2013) “Attribute-based Encryption with Verifiable Outsourced Decryption.” *IEEE Transactions on Information Forensics Security* **8**(8):1343-1354.
- [50] B. Qin, R. H. Deng, S. Liu, S. Ma. (2015) “Attribute-based Encryption with Efficient Verifiable Outsourced Decryption.” *IEEE Transactions on Information Forensics Security* **10**(7):1384-1394.
- [51] X. Mao, J. Lai, Q. Mei, K. Chen, J. Weng, “Generic and Efficient Constructions of Attribute-based Encryption with Verifiable Outsourced Decryption” *IEEE Transactions on Dependable and Secure Computing*, published online, DOI: **10.1109/TDSC.2015.2423669**.